

SOUTHEAST UNIVERSITY

Department of Computer Science and Engineering

CSE472: WEB AND INTERNET PROGRAMMING LAB

Lab Report 03

JavaScript Foundations and Simple Interaction

Student Workshop Registration System

Project Stage 03: Making the webpage interactive with JavaScript

Student information	Details
Student name	Rakibul Hasan
Student ID	REPLACE_STUDENT_ID
Section	REPLACE_SECTION
Department	Computer Science and Engineering
Submission date	08 September 2026
GitHub repository	github.com/USERNAME/cse472-web-lab-STUDENTID
Lab 03 folder	labs/lab-03-javascript/

Before submission: Replace the highlighted Student ID and Section, then replace the GitHub URL with the actual repository link.

2. Objectives

The purpose of this laboratory was to extend the Student Workshop Registration System from a static HTML and CSS page into an interactive webpage. The specific objectives were to:

- connect an external JavaScript file to the existing webpage;
- use a variable to store the number of available workshop seats;
- apply a basic `if...else` condition to select the correct seat message;
- create functions and call them from HTML buttons using `onclick`;
- select elements with `document.getElementById()`, read an input with `.value`, and update page text with `.textContent`; and
- implement and test one small independent improvement while keeping the code within the concepts introduced in Lab 03.

3. Tools Used

Table 1: Tools used to implement and verify Lab 03

Tool	Purpose
Visual Studio Code	Created and edited <code>index.html</code> , <code>css/style.css</code> , and <code>js/script.js</code> .
Modern web browser	Opened the page, activated buttons, entered a name, and observed text changes.
Browser Developer Tools	Checked element IDs, function connections, and JavaScript errors during debugging.
Node.js	Performed repeatable JavaScript interaction checks against controlled page elements.
Git and GitHub	Versioned the work and stored it in the required course repository structure.
LaTeX	Prepared the report with structured sections, figures, tables, links, and captions.

4. Work Completed

The Lab 02 layout was retained as a responsive registration webpage, while all required behaviour was placed in the new external file `js/script.js`. Figure 1 shows the completed workshop page structure.

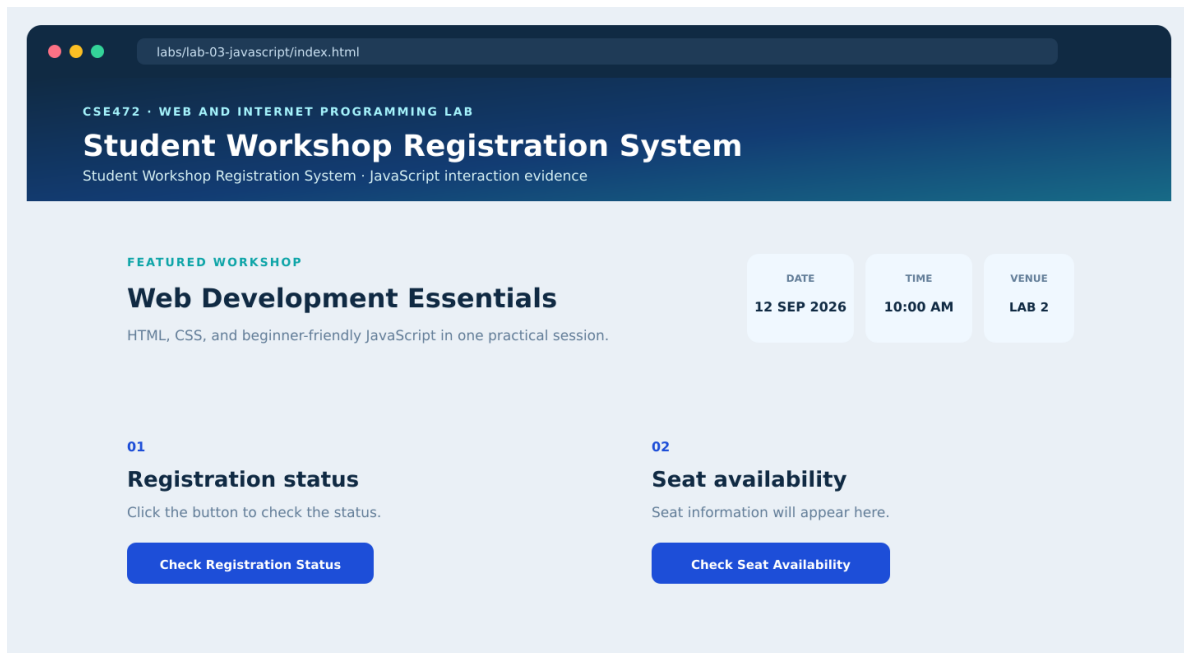


Figure 1: Completed Student Workshop Registration System interface.

Registration status interaction

The *Check Registration Status* button calls `checkRegistration()`. The function selects `registrationStatus` and replaces its original instruction with “Registration is currently open.”

Seat availability interaction

The variable `availableSeats` initially stores 12. The `checkSeats()` function uses `if...else`: a positive value displays the remaining seat count, while zero displays the full-workshop message.

Personal greeting interaction

The `showGreeting()` function reads the current value of `studentName`. It joins the typed name with a welcome message and writes the result into `greetingMessage`.

Independent improvement

A *Show Venue* button was added as the independent improvement. It calls `showVenue()` and changes the `venueMessage` paragraph to a short venue reminder. This feature uses only a function, `onclick`, `getElementById()`, and `textContent`.

5. Implementation Evidence

The selected evidence below focuses on the required connections and visible interaction results. The code images come directly from the submitted source files, and the output states were compared against the executed JavaScript tests.

```
js/script.js

JavaScript interactions
01 // Store the number of seats currently available for the workshop.
02 let availableSeats = 12;
03
04 // Display the current registration status on the webpage.
05 function checkRegistration() {
06     let message = document.getElementById("registrationStatus");
07     message.textContent = "Registration is currently open.";
08 }
09
10 // Use a simple decision to show whether seats are available.
11 function checkSeats() {
12     let message = document.getElementById("seatMessage");
13
14     if (availableSeats > 0) {
15         message.textContent = "Seats are available. Remaining seats: " + availableSeats;
16     } else {
17         message.textContent = "Sorry, no seats are available.";
18     }
19 }
20
21
22 // Read the student's typed name and show a personalised greeting.
23 function showGreeting() {
24     let name = document.getElementById("studentName").value;
25     let output = document.getElementById("greetingMessage");
26
27     output.textContent = "Welcome, " + name + "!";
28 }
29
30 // Independent improvement: display a short workshop venue reminder.
31 function showVenue() {
32     let message = document.getElementById("venueMessage");
33     message.textContent = "Venue: Computer Lab 2, Southeast University.";
34 }
35
```

Figure 2: Important sections of the completed external JavaScript file.

```
index.html (selected lines)

Required HTML connections
47 <p id="registrationStatus" class="result-message" aria-live="polite">Click the button to check the status.</p>
48 <button type="button" onclick="checkRegistration()">Check Registration Status</button>
54 <p id="seatMessage" class="result-message" aria-live="polite">Seat information will appear here.</p>
55 <button type="button" onclick="checkSeats()">Check Seat Availability</button>
70 <input type="text" id="studentName" name="studentName" placeholder="Enter your full name" required>
95 <button type="button" onclick="showGreeting()">Show Greeting</button>
97 <p id="greetingMessage" class="result-message wide" aria-live="polite">Your personalised greeting will appear here.</p>
105 <p id="venueMessage" class="result-message" aria-live="polite">Click the button to display the workshop venue.</p>
107 <button type="button" onclick="showVenue()">Show Venue</button>
115 <script src="js/script.js"></script>
```

Figure 3: Selected HTML IDs, onclick calls, and external script connection.

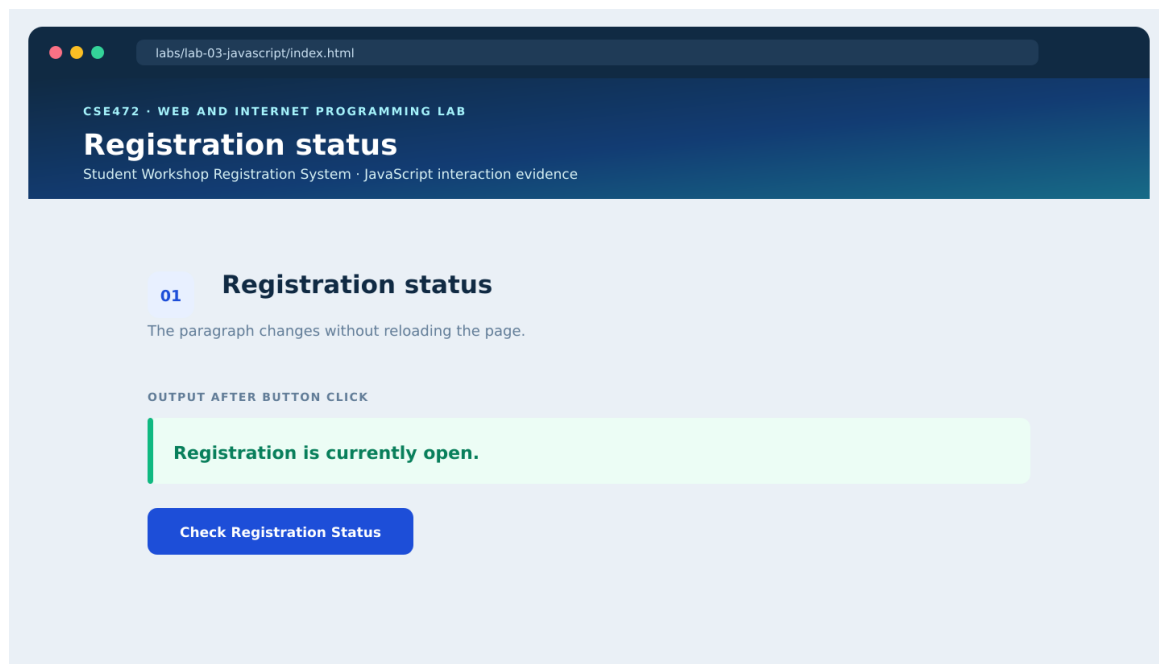


Figure 4: Output after clicking *Check Registration Status*.

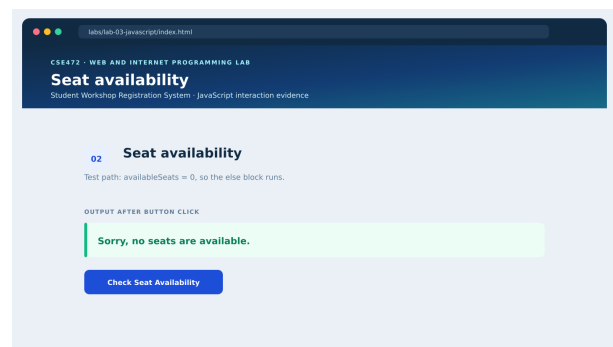
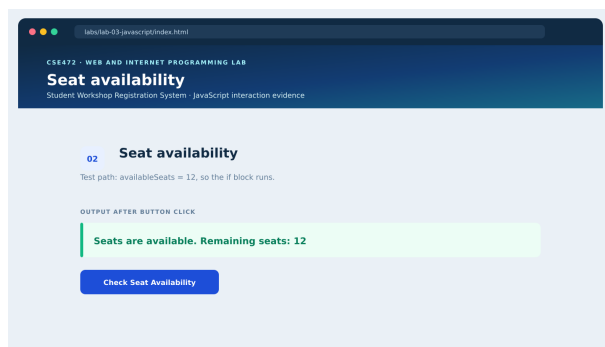


Figure 5: Seat checker with `availableSeats = 12`. Figure 6: Seat checker with `availableSeats = 0`.

Figures 5 and 6 demonstrate both branches of the required condition. The original value was restored to 12 after the zero-seat test.

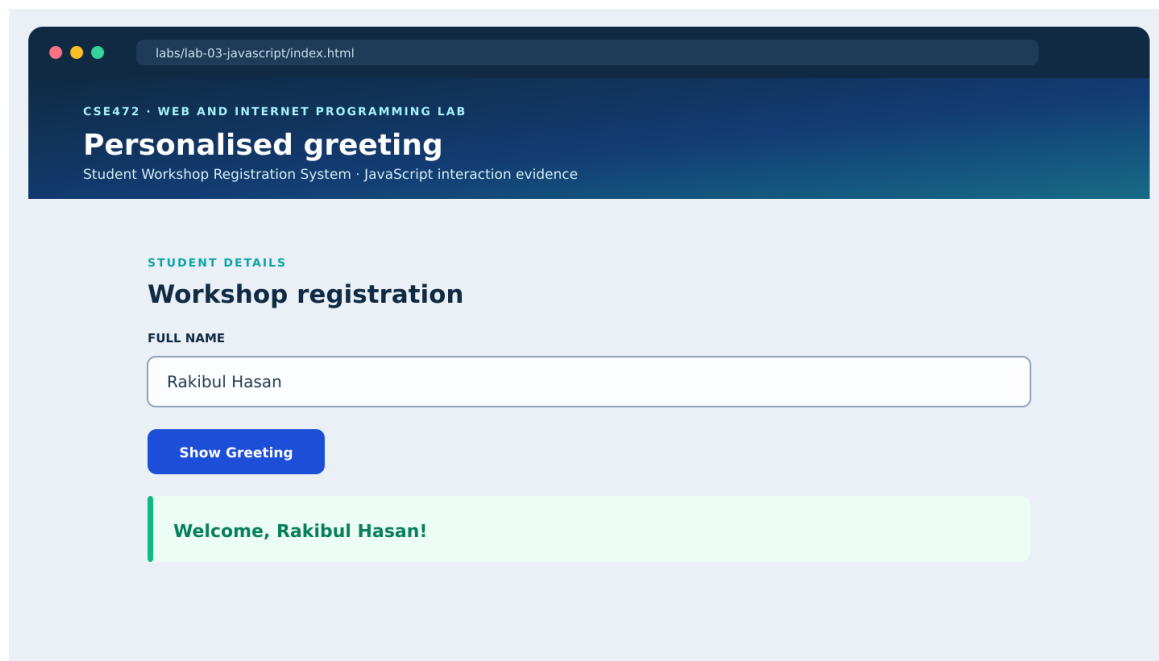


Figure 7: Personalised greeting after reading “Rakibul Hasan” from the Full Name input.

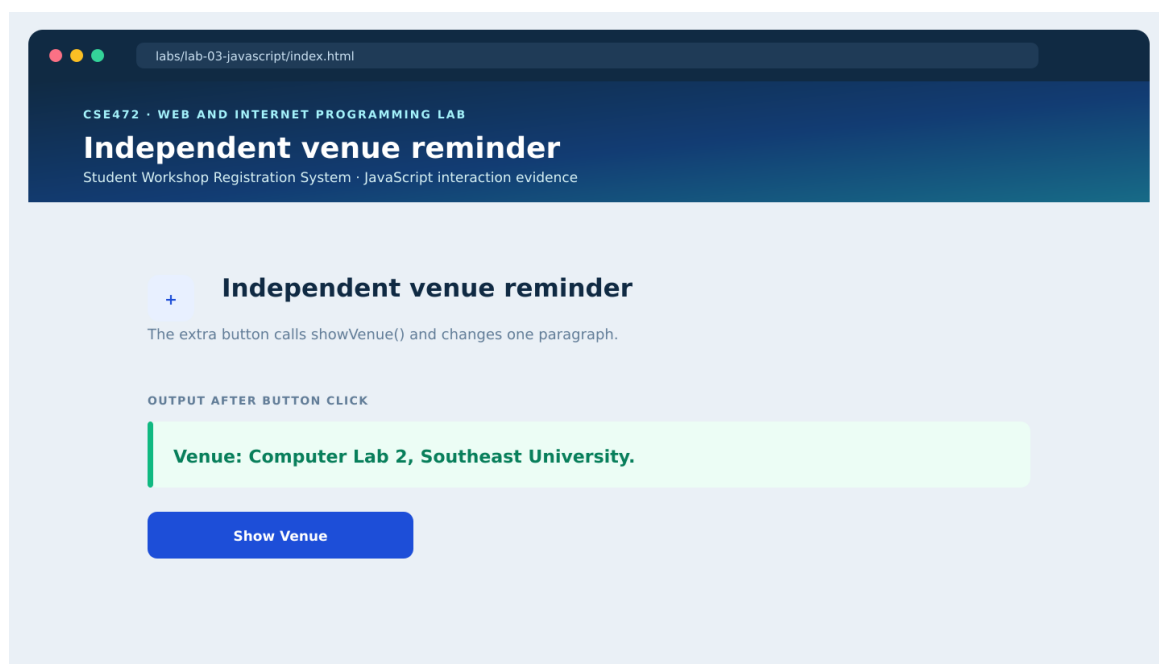


Figure 8: Independent venue-reminder interaction.

6. Key JavaScript Explanation

Table 2: Lab 03 JavaScript concepts explained using the implemented code

Concept	Example	Explanation
Variable	<code>let availableSeats = 12;</code>	Stores a value under a meaningful name so it can be used later.
<code>if...else</code>	<code>if (availableSeats > 0)</code>	Chooses between the available-seat message and the no-seat message.
Function	<code>function checkSeats() {...}</code>	Groups related instructions under a name. Creating it defines the instructions; calling it runs them.
<code>onclick</code>	<code>onclick="checkSeats()"</code>	Calls the named JavaScript function when the user clicks the HTML button.
<code>getElementById()</code>	<code>getElementById("seatMessage")</code>	Finds the unique HTML element whose <code>id</code> is exactly <code>seatMessage</code> . The letter case must match.
<code>textContent</code>	<code>message.textContent = ...</code>	Replaces the visible text inside the selected paragraph.
<code>value</code> <code>.value</code>	<code>getElementById("studentName")</code> Reads the text currently typed into the Full Name input.	

For example, clicking the seat button calls `checkSeats()`. The function finds the `seatMessage` paragraph, checks `availableSeats`, and changes the paragraph's `textContent`. The page does not need to reload.

7. Testing and Corrections

Each required interaction was tested separately. A repeatable JavaScript check also confirmed that the expected HTML connections were present and that every function named by `onclick` existed. All seven automated checks passed.

Table 3: Functional testing results

Test	Input or action	Expected and observed result	Status
Script connection	Load <code>index.html</code>	External file path is <code>js/script.js</code> ; all functions execute.	PASS
Registration status	Click status button	Paragraph becomes "Registration is currently open."	PASS
Seat availability, true path	Set seats to 12; click button	Remaining-seat message displays 12.	PASS
Seat availability, false path	Set seats to 0; click button	"Sorry, no seats are available." is displayed.	PASS
Personal greeting	Enter "Rakibul Hasan"; click greeting	"Welcome, Rakibul Hasan!" is displayed.	PASS
Independent improvement	Click <i>Show Venue</i>	Computer Lab 2 venue reminder appears.	PASS
Function-name check	Inspect every <code>onclick</code> call	No missing or incorrectly cased function names.	PASS

Error identified and corrected

The required debugging check showed why exact letter case matters. A deliberately mismatched lookup such as `registrationstatus` does not find the HTML element named `registrationStatus`. The lookup was corrected so the HTML id and the argument passed to `getElementById()` match exactly. The same case check was applied to every function name used by `onclick`.

The detailed machine-readable result is included in the submission package as `evidence/automated_test_results.txt`.

8. GitHub Link

The source code is organised in the required repository location:

Repository: <https://github.com/USERNAME/cse472-web-lab-STUDENTID>
Required folder: `labs/lab-03-javascript/`

Final edit required: Replace the placeholder URL above with the actual personal GitHub repository link before submitting the PDF.

The suggested commit message is: `Complete Lab 03 JavaScript interactions.`

9. Learning Reflection

1. What is one clear difference between HTML and JavaScript?

HTML defines the structure and content of the webpage, such as headings, inputs, paragraphs, and buttons. JavaScript adds behaviour, such as responding to a click, reading typed information, making a decision, and changing visible text.

2. Which JavaScript concept was most useful in this laboratory? Why?

The combination of functions and `getElementById()` was most useful. A function keeps each interaction organised, while `getElementById()` connects that function to a specific part of the webpage. Together they make the button actions clear and reusable.

3. Describe one error you faced and how you fixed it.

A case mismatch between `registrationstatus` and `registrationStatus` prevented the intended element from being found during the debugging check. I fixed it by matching the JavaScript lookup exactly to the HTML id, then checked all remaining IDs and function names in the same way.

4. What simple JavaScript improvement did you add independently?

I added a *Show Venue* button. When clicked, it calls `showVenue()`, finds the `venueMessage` paragraph, and updates the paragraph with the workshop venue.

10. AI Assistance Declaration

I used **ChatGPT** to help organise the report, check the JavaScript logic, generate repeatable test evidence, and improve the clarity of the written explanations. I checked the generated code by reviewing each function, confirming the HTML IDs and `onclick` names, running both paths of the seat condition, comparing every output with its expected text, and keeping the implementation within the concepts taught in Lab 03.

A. Practice Set Answers

Predict the result

1. The value of `remaining` is 5, because $8 - 3 = 5$.
2. The browser displays `Full`, because `availableSeats` is zero and the condition `availableSeats > 0` is false.

Small coding tasks

Listing 1: Practice variables and functions

```
1 let courseName = "CSE472";
2 alert(courseName);
3
4 function showCourse() {
5     alert("Web and Internet Programming Lab");
6 }
7
8 function showNotice() {
9     let notice = document.getElementById("notice");
10    notice.textContent = "Lab starts at 9:00 AM.";
11 }
12
13 function showStudentId() {
14     let id = document.getElementById("studentId").value;
15     let output = document.getElementById("idOutput");
16     output.textContent = "Student ID: " + id;
17 }
```

Listing 2: Practice HTML connections

```
1 <button type="button" onclick="showCourse()">Show Course</button>
2
3 <p id="notice">Waiting for the lab notice.</p>
4 <button type="button" onclick="showNotice()">Show Notice</button>
5
6 <input type="text" id="studentId">
7 <button type="button" onclick="showStudentId()">Show Student ID</button>
8 <p id="idOutput"></p>
```

Concept explanations

- JavaScript adds behaviour and interaction to a webpage.
- An HTML id must match `getElementById()` exactly so the browser can find the intended element.
- Creating a function defines its instructions; calling the function with parentheses runs those instructions.
- `.value` reads the current content of an input field.
- `.textContent` reads or changes the text inside an HTML element.

B. Project Structure and Submission Checklist

Required project structure

```
1 labs/
2 '-- lab-03-javascript/
3   |-- index.html
4   |-- css/
5   |   '-- style.css
6   |-- js/
7   |   '-- script.js
8   '-- assets/
9       '-- images/
```

Final checks

Done	Checklist item
☒	The report has the correct Lab 03 title and all ten required sections.
☒	The code uses the required external JavaScript file and exact project folder.
☒	Selected code and interaction evidence include captions.
☒	Registration status, both seat paths, greeting, and independent button were tested.
☒	Reflection answers and the AI assistance declaration are included.
☐	Replace Student ID, Section, and GitHub URL; then rename the PDF to CSE472_Lab03_StudentID.pdf.